CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

# 9.2 Potential Difference & Power

**Syllabus Section:** 9. Electricity **Paper Type:** Multiple Choice (Paper 1) **Total:** 18 Questions (18 Marks)

## Question 1 Easy [1 mark]

What describes the electric potential difference between two points in a wire that carries a current?

A the force required to move a unit positive charge between the points

B the ratio of the energy dissipated between the points to the current

C the ratio of the power dissipated between the points to the current

D the ratio of the power dissipated between the points to the charge moved

### Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Power is equal to $P = VI$
  - So, potential difference is equal to $V = \frac{P}{I}$
- This equation tells us that potential difference between two points in a wire is equal to the power dissipated between the points divided by the current flowing in the wire
  - This is also known as the ratio of these two quantities

**A** is incorrect because this describes the work done in moving a charge, which is not the same as the power dissipated

**B** is incorrect because power and energy are different quantities

**D** is incorrect because current and charge are different quantities

**Question 2** Easy [1 mark]

A filament lamp has a resistance of 180 Ω when the current in it is 500 mA.

What is the power transformed in the lamp?

**A** 45 W

**B** 50 W

**C** 90 W

**D** 1400 W

**Mark Scheme & Explanation:**

*The correct answer is **A** because:*

- Power is equal to $P = VI$
- Potential difference, current and resistance are related by the equation $V = IR$
- We are given current and resistance, so we need to eliminate *V*
- Combining the two equations gives
  - $P = I \quad RI = I^{\ 2}\ R$
- So, $P = 500 \times 10^{-3^{\ 2}} \times 180 = 45\text{W}$
- Therefore, the power transformed in the lamp is 45 W

**Question 3** Easy [1 mark]

A 12 V battery is charged for 20 minutes by connecting it to a source of electromotive force (e.m.f.). The battery is supplied with $7.2 \times 10^4$ J of energy in this time.

How much charge flows into the battery?

**A** 5.0 C

**B** 60 C

**C** 100 C

**D** 6000 C

**Mark Scheme & Explanation:**

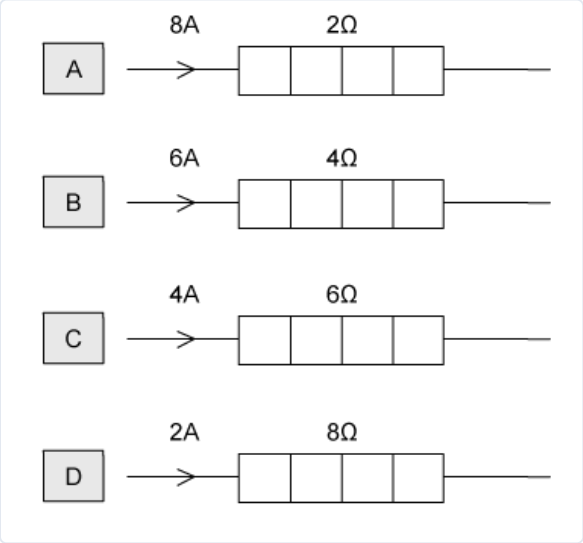
*The correct answer is **D** because:*

- Potential difference is defined as the amount of electrical energy transferred between two points per unit of charge
  - In equation form this can be expressed as $V = \frac{E}{Q}$
  - Rearranging for charge Q gives $Q = \frac{E}{V}$
- Therefore, the charge that flows is $Q = \frac{7.2 \times 10^4}{12} = 6000 \quad \text{C}$

**Question 4** Easy [1 mark]

The diagram shows four heaters and the current in each.

Which heater has the greatest power dissipation ?




A

B

C

D

**Mark Scheme & Explanation:**

*The correct answer is **B** because:*

- Power is equal to $P = VI$
- Potential difference, current and resistance are related by the equation $V = IR$
- We are given current and resistance, so we need to eliminate *V*
- Combining the two equations gives
  - $P = IRI = I^2 R$
- Let's consider option B
  - Current *I* = 6 A, Resistance *R* = 4 Ω
  - $P = 6^2 \times 4 = 144 \text{ W}$
- This is the highest power dissipation out of all the options (see other calculations below)

**A** is incorrect because $P = I^2 R = 8^2 \times 2 = 128 \text{ W}$

**C** is incorrect because $P = I^2 R = 4^2 \times 6 = 96 \text{ W}$

**D** is incorrect because $P = I^2 \quad R = 2^2 \times 8 = 32 \quad \text{W}$

## Question 5 Easy [1 mark]

What is the correct definition for potential difference?

**A** Energy transferred per unit charge

**B** Energy transferred per unit current

**C** Power transfer per unit charge

**D** Work done per unit current

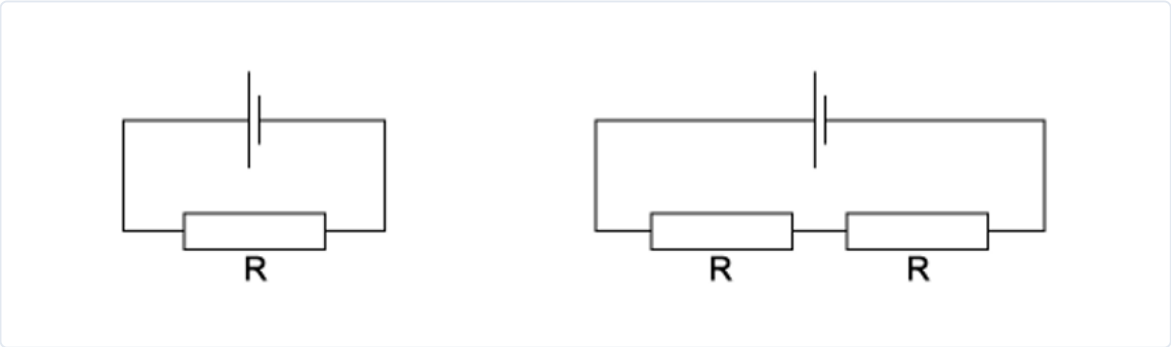
### Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Potential difference is the energy transferred per unit charge, measured in volts (V) which is equivalent to joules per coulomb (J $C^{-1}$)

**Question 6** Medium [1 mark]

The diagrams show two different circuits.




The cells in each circuit have the same electromotive force and zero internal resistance. The three resistors each have the same resistance *R*.

In the circuit on the left, the power dissipated in the resistor is *P*.

What is the total power dissipated in the circuit on the right?

A $\frac{P}{4}$

B $\frac{P}{2}$

C P

D 2P

**Mark Scheme & Explanation:**

*The correct answer is **B** because:*

- Power is equal to $P = VI$
- Current is equal to $I = \frac{V}{R}$
- Combining the two equations gives
  - $P = V\frac{V}{R} = \frac{V^2}{R}$
- so, power dissipated in the circuit on the right is $P = \frac{V^2}{R}$
- The resistance in the circuit on the left is 2R
- So, power dissipated in the circuit on the right is $\frac{V^2}{2R} = \frac{1}{2}\frac{V^2}{R} = \frac{P}{2}$

- Therefore, the power dissipated by the circuit on the right is $\frac{P}{2}$

## Question 7 Medium [1 mark]

The potential difference across a component in a circuit is 2.0 V.

How many electrons must flow through this component in order for it to be supplied with 4.8 J of energy?

A $2.6 \times 10^{18}$

B $1.5 \times 10^{19}$

C $3.0 \times 10^{19}$

D $6.0 \times 10^{19}$

**Mark Scheme & Explanation:**

*The correct answer is **B** because:*

- Potential difference is defined as the amount of electrical energy transferred between two points per unit of charge
  - In equation form this can be expressed as $V = \frac{E}{Q}$
  - Rearranging for charge Q gives $Q = \frac{E}{V}$
  - So, charge is equal to $Q = \frac{4.8}{2.0} = 2.4 \text{ C}$
- The total charge is equal to $Q = ne$
  - Where *n* is the number of electrons and *e* is the charge of one electron
- So, the number of electrons is equal to $n = \frac{Q}{e} = \frac{24}{1.6 \times 10^{-19}} = 1.5 \times 10^{19}$

**Question 8** Medium [1 mark]

Two lamps are connected in series to a 250V power supply. One lamp is rated 240V, 60W and the other is rated 10V, 2.5W.

Which statement most accurately describes what happens ?

A both lamps light at less than their normal brightness

B both lamps light normally

C only the 60 W lamp lights

D the 10 V lamp blows

**Mark Scheme & Explanation:**

*The correct answer is **B** because:*

- One lamp is rated 240 V, 60 W, this means that for the lamp to operate normally, there should be a potential difference (p.d.) of 240 V across it
- Electrical power is equal to $P = VI$
  - So, the current flowing through this lamp $= \frac{P}{V} = \frac{60}{240} = 0.25 \quad \text{A}$
- The other lamp is rated 10 V, 2.5 W. This lamp operates normally when there is a p.d. of 10 V across it
  - Current flowing through the lamp $= \frac{P}{V} = \frac{2.5}{10} = 0.25 \quad \text{A}$
- So, for normal operation, the same current of 0.25 A is required for both lamps. Since the lamps are connected in series, the same current would flow
  - So, both lamps light at their normal brightness
- In the same way, the sum of p.d. across the 2 lamps is equal to 240 + 10 = 250 V

**Question 9** Medium [1 mark]

A power cable X has resistance *R* and carries current I.

A second cable Y has resistance 2*R* and carries current $\frac{I}{2}$

What is the ratio $\frac{power\ \ dissipated\ \ in\ \ Y}{power\ \ dissipated\ \ in\ \ X}$ ?

A $\frac{1}{4}$

B $\frac{1}{2}$

C 2

D 4

**Mark Scheme & Explanation:**

*The correct answer is **B** because:*

- Power is equal to $P = VI$
- Potential difference is equal to $V = IR$
- Combining the two equations gives
  - $P = IRI = I^2R$
- The power dissipated in X is $= I^2R$
- The power dissipated in Y is $= \frac{1}{2}^2 2R = \frac{2I^2R}{4} = \frac{I^2R}{2}$
  - Therefore, the ratio $\frac{power\ \ dissipated\ \ in\ \ Y}{power\ \ dissipated\ \ in\ \ X} = \frac{I^2R}{2} \times \frac{1}{I^2R} = \frac{1}{2}$

**Question 10** Medium [1 mark]

In a simple electrical circuit, the current in a resistor is measured as (2.50 ± 0.05) mA. The resistor is marked as having a value of 4.7Ω ± 2 %.

If these values were used to calculate the power dissipated in the resistor, what would be the percentage uncertainty in the value obtained?

A 2 %

B 4 %

C 6 %

D 8 %

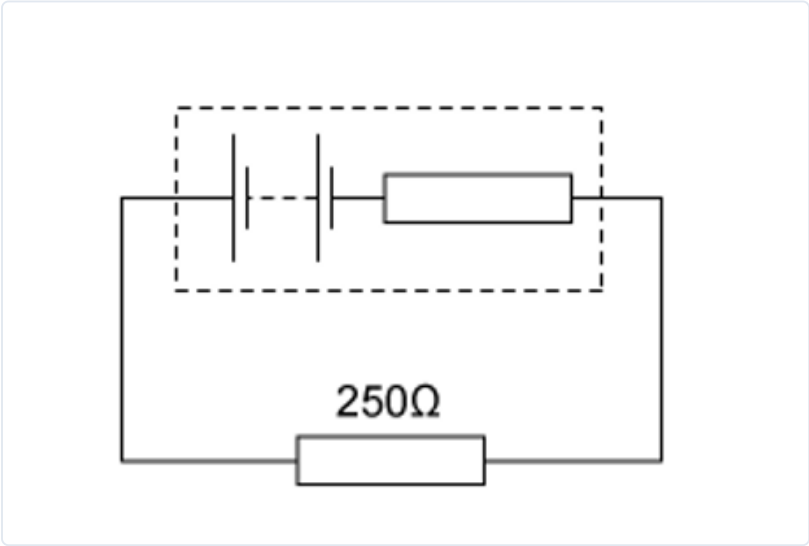
**Mark Scheme & Explanation:**

*The correct answer is **C** because:*

- Power is equal to $P = VI$
- Potential difference is equal to $V = IR$
- Combining the two equations gives
  - $P = IRI = I^2R$
- First, we must convert the uncertainty in current to a percentage uncertainty so we can add the percentage uncertainties together
  - Current $I = 2.5 \text{ A}$
  - Uncertainty in current $\Delta I = 0.05 \text{ A}$
- So, the percentage uncertainty in current $= \frac{\Delta I}{I} \times 100 \text{ \%} = \frac{0.05}{2.5} \times 100 = 2 \text{ \%}$
- Since the power dissipated is given by $P = I^2 R$, the percentage uncertainty in power is $= 2 \times \%\ uncertainity\ in\ I + \%\ uncertainity\ in\ R$
  - Therefore, $\%\ uncertainity\ in\ P = 2 \times 2 \%\ + 2\% = 4 + 2 = 6 \%$

## Question 11 Medium [1 mark]

A battery, with a constant internal resistance, is connected to a resistor of resistance 250 Ω, as shown.




The current in the resistor is 40 mA for a time of 60 s. During this time 6.0 J of energy is lost in the internal resistance.

What is the energy supplied to the external resistor during the 60 s and the e.m.f of the battery ?

| | Energy / J | E. m. f / V |
|---|---|---|
| **A** | 2.4 | 2.4 |
| **B** | 2.4 | 7.5 |
| **C** | 24 | 10.0 |
| **D** | 24 | 12.5 |

A

B

C

D

### Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Power is equal to $P = VI = IRI = I^2R$
- The energy transferred in external resistor is equal to $E = Pt = I^2 R t$

- Current $I = 40 \quad \text{mA} = 40 \times 10^{-3} \quad \text{A}$
- Resistance $R = 250 \quad \Omega$
- Time $t = 60 \quad \text{s}$

- Energy transferred to the resistor $E = 40 \times 10^{-3}{}^{2} \times 250 \times 60 = 24 \quad \text{J}$
  - Therefore, since 6 J of energy is lost in internal resistance, the battery must supply a total of 6 + 24 = 30 J of energy to the whole circuit
- The energy supplied by battery is $E = QV$
- Where *Q* is the charge and *V* is the e. m .f of the battery.
- Charge transferred, $Q = I \quad t = 40 \times 10^{-3} \times 60 = 2.4 \quad \text{C}$
  - So, e. m. f of battery, $V = \frac{E}{Q} = \frac{30}{2.4} = 12.5 \quad \text{V}$

**Question 12** Medium [1 mark]

The charge that an electric battery can deliver is specified in ampere-hours.

For example, a battery of capacity 40 ampere-hours could supply, when fully charged, 0.2 A for 200 hours.

What is the maximum energy that a fully charged 12 V, 40 ampere-hour battery could supply ?

**A** 1.7 kJ

**B** 29 kJ

**C** 1.7 MJ

**D** 29 MJ

**Mark Scheme & Explanation:**

*The correct answer is **C** because:*

- Electrical power is equal to $P = VI$
- Power is also equal to $P = \frac{E}{t}$
  - Combining the two equations, $P = \frac{E}{t} = VI$
  - So, energy is equal to $E = VI \quad t$
- The battery is described as '12 V, 40 ampere-hour', this means that the e. m .f. of the battery is $V$ = 12 V
- From the example given, 40 ampere-hours means that a current of 0.2 A can be supplied for 200 hours or 0.2 A × 200 h = 40 A h
- We need to convert this to SI units
- We know that 1 hour = 3600 s, so, 40 ampere-hours = 40 × 3600 ampere-seconds
- This quantity represents current × time or $I \quad t$
  - Energy to $E = VI \quad t = 12 \times 40 \times 3600 = 1 \quad 728 \quad 000 \quad \text{J}$
- Therefore, the maximum energy that can be supplied is 1.7 MJ

**Question 13** Hard [1 mark]

The amount of energy transferred when 10 C of charge passes through a p. d. of 20 V is the same as the energy needed to raise a 2 kg mass through a distance *x*.

Take gravitational field strength to be 10 N $kg^{-1}$.

What is the value of *x*?

**A** 0.1 m

**B** 1.0 m

**C** 10 m

**D** 100 m

**Mark Scheme & Explanation:**

*The correct answer is **C** because:*

- Potential difference is defined as the amount of electrical energy transferred between two points per unit of charge
  - In equation form this can be expressed as $V = \frac{E}{Q}$
  - Rearranging for energy gives $E = VQ$
- So, the amount of energy transferred $= VQ = 20 \times 10 = 200 \text{ J}$
- Gravitational potential energy is equal to $= mgh$
  - Where *h* is the vertical distance moved by the mass, $h = x$
- We can equate this with the energy transferred by the charge:
  - $mgx = 200 \text{ J}$
  - $x = \frac{200}{2 \times 10} = 10 \text{ m}$
- Therefore, the charge is raised by 10 m

**Question 14** Hard [1 mark]

The ratings of two lamps are as follows:

| | |
|---|---|
| **X** | 230 V, 100 W |
| **Y** | 230 V, 60 W |

During normal operation, which of the lamps will have larger current and resistance?

**A** $I_X > I_Y \quad and \quad R_X > R_Y$

**B** $I_X > I_Y \quad and \quad R_X < R_Y$

**C** $I_X < I_Y \quad and \quad R_X > R_Y$

**D** $I_X < I_Y \quad and \quad R_X < R_Y$

**Mark Scheme & Explanation:**

*The correct answer is **B** because:*

- Power is equal to $P = VI$
- Current is equal to $I = \frac{V}{R}$
- Combining the two equations gives
  - $P = V\frac{V}{R} = \frac{V^2}{R}$
- For the same value of $V$:
  - $P \ \alpha \ I$ and $P \ \alpha \ \frac{1}{R}$
- X has a higher power rating than Y, so it will have a greater current
  - $I_X > I_Y$
- X has a higher power rating than Y, so it will have a smaller resistance
  - $R_X < R_Y$

**Question 15** Hard [1 mark]

Two lightbulbs, **X** and **Y**, are in parallel and are attached to a cell which has a potential difference of 75 V. Lightbulb **X** has a resistance of 350 Ω and Lightbulb **Y** has a resistance of 500 Ω. The power of Lightbulbs **X** and **Y** are $P_X$ and $P_Y$ respectively.

What is the value of the ratio $\frac{P_X}{P_Y}$?

**A** $\frac{1}{2}$

**B** $\frac{7}{10}$

**C** $\frac{10}{7}$

**D** 2

**Mark Scheme & Explanation:**

*The correct answer is **C** because:*

- The known quantities are
  - Resistance of Lightbulb **X**, $R_X$ = 350 Ω
  - Resistance of Lightbulb **Y**, $R_Y$ = 500 Ω
- The power of the lightbulbs can be found using
  - $P = \frac{V^2}{R}$
- The potential difference for each light bulb will be the same as they are in parallel, therefore
  - $P \propto \frac{1}{R}$
- The power of Lightbulbs **X** and **Y** are
  - $P_X \propto \frac{1}{350}$
  - $P_Y \propto \frac{1}{500}$
- The value of the ratio of $P_X$ and $P_Y$ is
  - $\frac{P_X}{P_Y} = \frac{1}{350} \times \frac{500}{1} = \frac{500}{350}$
  - $\frac{P_X}{P_Y} = \frac{10}{7}$
- This is option **C**

**Question 16** Hard [1 mark]

What is the effect on the current and power of a fixed resistor when the voltage across the resistor doubles?

| | Current | Power |
|---|---|---|
| **A.** | Quadruples | Doubles |
| **B.** | Doubles | Doubles |
| **C.** | Doubles | Quadruples |
| **D.** | Halves | Doubles |

A

B

C

D

**Mark Scheme & Explanation:**

*The correct answer is **C** because:*

- $V = IR$
- $P = \frac{V^2}{R}$
- A fixed resistor means that the resistance does not change, therefore
  - $V \propto I$
  - $P \propto V^2$
- When the voltage doubles, the current also doubles
  - $2V \propto 2I$
- When the voltage doubles, the power quadruples
  - $2V^2 = 4V^2$
  - $4P \propto 2V^2$
- This is option **C**

## Question 17 Hard [1 mark]

Two resistors, **X** and **Y**, are in series with a cell that has a potential difference of *V*. The resistance of **X** and **Y** is *R* and $\frac{R}{2}$ respectively.

Given that the charge through both resistors is 6 C, what is the power and work done of Resistor **X**?

| | Power | Work done |
|---|---|---|
| **A.** | 2V | $\frac{8V^2}{9R}$ |
| **B.** | 3V | $\frac{V^2}{2R}$ |
| **C.** | 4V | $\frac{4V^2}{9R}$ |
| **D.** | 6V | $\frac{V^2}{R}$ |

A

B

C

D

### Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The known quantities are
  - The charge through the resistors, $Q$ = 6 C
  - Resistance of **X**, $R_X = R$
  - Resistance of **Y**, $R_Y = \frac{R}{2}$
  - Potential difference in, $V_{IN} = V$
- The potential difference of Resistor **X** can be calculated using
  - $V_{OUT} = \frac{R_1}{R_1 + R_2} V_{IN}$
- Resistor **X** ($R_1$) has a resistance of $R$, therefore
  - $V_{OUT} = \frac{R}{R + \frac{R}{2}} V$
  - $V_{OUT} = \frac{R}{\frac{3R}{2}} V$

- $V_{OUT} = \frac{2}{3}V$

- The work done by Resistor **X** is
  - $W = QV$
  - $W = 6 \times \frac{2}{3}V = 4V$
- The power of Resistor **X** is
  - $P = \frac{V^2}{R}$
  - $P = \frac{\frac{2}{3}V^2}{R}$
  - $P = \frac{4V^2}{9R}$
- This is option **C**

**Question 18** Hard [1 mark]

A battery with an electromotive force (e.m.f.) of 240 V and an internal resistance of 2.0 Ω is connected to a heating element. The element draws a current of 8.7 A.

The power dissipated by the heating element is the same as the power that would be dissipated by a 12 Ω resistor carrying a current of *I*.

What is the value of *I*?

**A** 1.3 A

**B** 4.1 A

**C** 13 A

**D** 130 A

**Mark Scheme & Explanation:**

*The correct answer is C*

- List the known quantities
  - E.m.f., $E = 240$ V
  - Internal resistance of battery, $r = 2.0$ Ω
  - Current through heating element, $I_H = 8.7$ A
  - Resistance, $R = 12$ Ω
- Calculate the potential difference across the heating element
  - When a current $I_H$ is drawn from a non-ideal source (one with internal resistance *r*), the terminal potential difference $V_H$ across the load is less than the e.m.f. $E$ due to the lost volts $I_H r$

$$V_H = E - I_H r$$

$$V_H = 240 - 8.7 \times 2.0$$

$$V_H = 222.6 \text{ V}$$

- Calculate the power dissipated by the heating element

$$P = V_H I_H$$

$$P = 222.6 \times 8.7 = 1937 \text{ W}$$

- Calculate the current through the resistor

$$P = I^2 R$$

$$I = \sqrt{\frac{P}{R}}$$

$$I = \sqrt{\frac{1937}{12}} = 12.7 \text{ A} = 13 \text{ A } 2 \text{ s.f.}$$